

Effect of FDC of Sitagliptin, Glimepiride and Metformin IR on post-prandial blood glucose in patients with T2DM: A post-hoc analysis of phase 3 study

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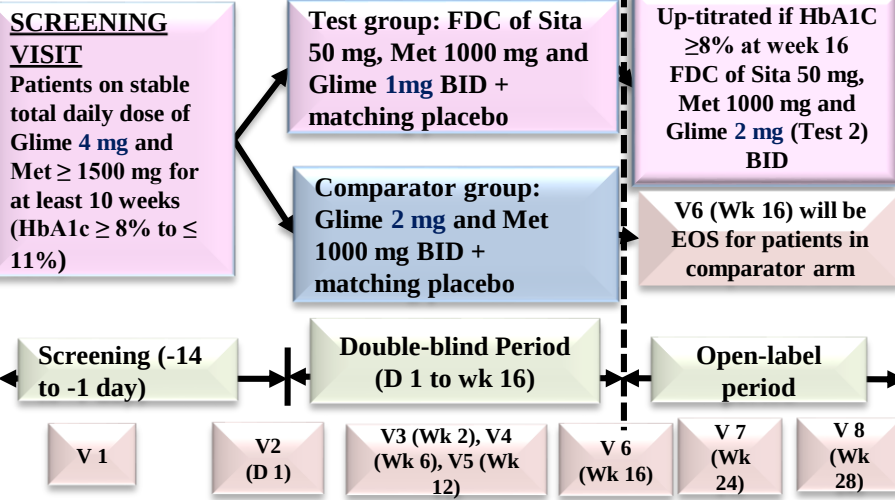
BACKGROUND

Importance of Postprandial Blood Glucose:

- Long-term glycemic targets cannot be achieved by targeting only fasting blood glucose (FBG) levels, and evidence supports the importance of postprandial blood glucose (PPBG) in glycemic control with regard to the development of complications in diabetes^{1,2}
- From the viewpoint of blood glucose control in diabetes, PPBG level is more relevant than FBG level to the development and progression of macroangiopathy
- The DECODE study has shown that accelerated postprandial excursion is strongly associated with occurrence of cardiovascular diseases³
- International Diabetes Federation (IDF) recommends target PPBG levels of <160 mg/dL⁴

METHODS

- This is a post-hoc analysis of a phase 3, randomized, double blind, active-controlled study followed by an open-label study for 12 weeks [CTRI/2021/11/038169]



BID: Twice daily; D: Day; EOS: End of study; FDC: Fixed dose combination; Glime: Glimepiride; HbA1c: Glycated hemoglobin; Met: Metformin; Sita: Sitagliptin; V: Visit; Wk: Week

- This post-hoc analysis evaluated control of postprandial blood glucose (PPBG) at Week 16, proportion of patients achieving mean PPBG <160 mg/dL, and the proportion of patients achieving both mean PPBG <160 mg/dL and HbA1c <7% at Week 16

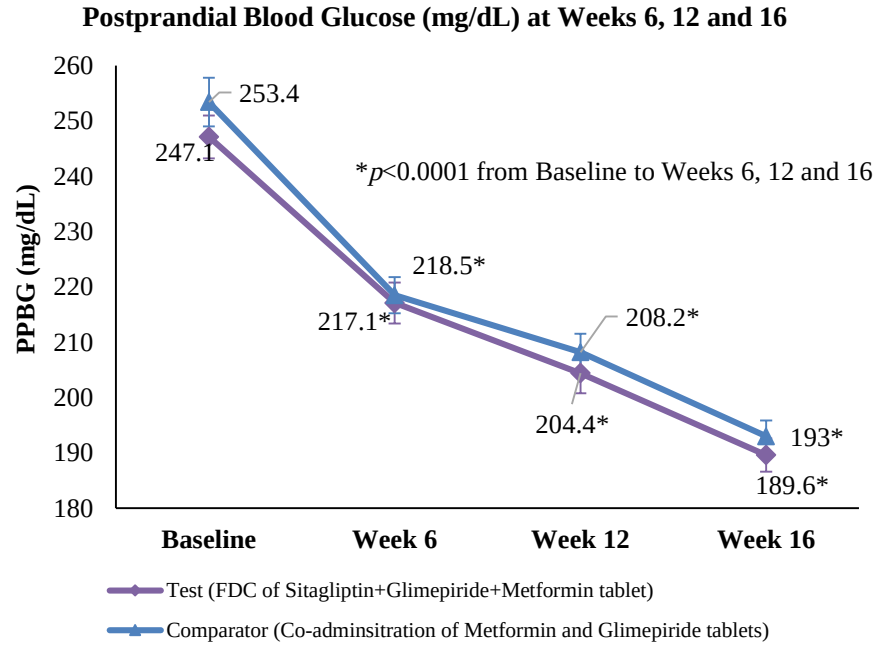
RESULTS

Baseline Demographics:

- A total of 392 patients (Test=190 and Comparator=202) were included in this post-hoc analysis
- The overall mean age was 47.29 ± 8.87years and the mean BMI was 25.73 ± 3.75 Kg/m²
- The overall baseline mean HbA1c (%) was 9.16 ± 0.75; baseline mean FBG was 180.6 ± 39.44 mg/dL; and baseline mean PPBG was 253.5 ± 58.51 mg/dL
- Baseline characteristics were similar in Test arm and Comparator arm

PPBG from Baseline to Week 16:

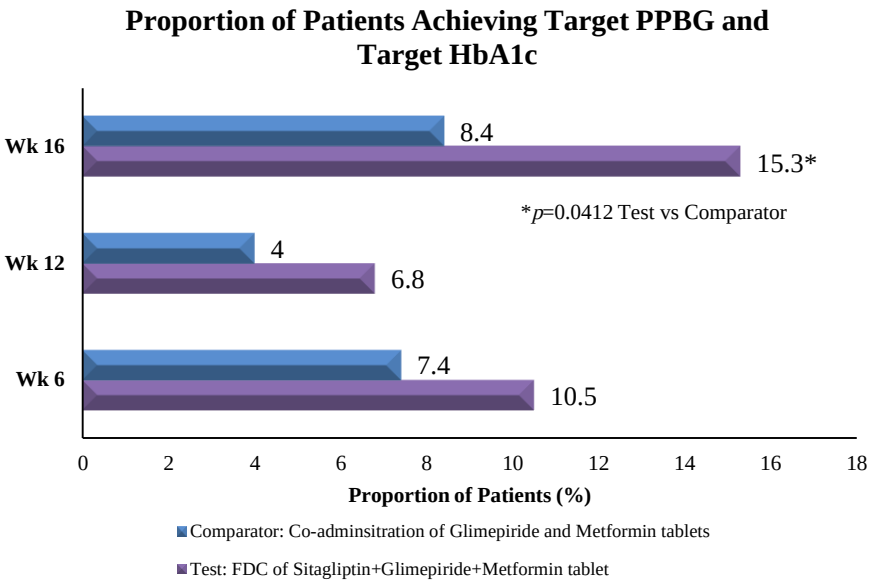
- At Week 16, in Test and Comparator arms, PPBG levels reduced from 247.1 mg/dL and 253.4 mg/dL to 189.6 mg/dL and 193 mg/dL, respectively



Proportion of patients achieving Mean PPBG <160 mg/dL:

- In Test arm versus Comparator arm, proportion of patients achieving mean PPBG <160 mg/dL at Weeks 6, 12 and 16 was 10.10% versus 7.40%, 11.80% versus 10.60% and 26.7% versus 24.7%, respectively
- Proportion of patients achieving mean PPBG <160 mg/dL was higher in test arm

Proportion of Patients Achieving PPBG <160 mg/dL and HbA1c <7.0%:



- Proportion of patients achieving both PPBG <160 mg/dL and HbA1c <7.0% statistically significant in Test arm as compared to Comparator arm at Week 16 (Test versus Comparator: 15.3% versus 8.4%, p=0.0412)

Adverse Events (AE) – Overall Summary:

	Test arm (N=190)		Comparator arm (N=202)		Overall (N=392)	
	n (%)	E	n (%)	E	n (%)	E
Any TEAEs	25 (13.2%)	31	30 (14.9%)	36	55 (14.0%)	67
SAEs	0	0	0	0	0	0
Drug-related TEAEs	12 (6.3%)	12	16 (7.9%)	16	28 (7.1%)	28
Severe TEAEs	0	0	1 (0.5%)	1	1 (0.3%)	1

Test: FDC of Sitagliptin+Glimepiride+Metformin (50+1+1000 mg) BID; Comparator: Glimepiride 2 mg + Metformin 1000 mg BID

BID: Twice daily; E: Number of events; FDC: Fixed dose combination; N: Total number of patients; n: number of patients; SAE: Serious AE; TEAE: Treatment emergent AE

- The most commonly observed adverse events (>1%) were diarrhea, vomiting, pyrexia and headache
- Out of 67 TEAEs, 58 TEAEs (29 each in Test and Comparator arms) were mild in nature; 8 TEAEs (Test=2 and Comparator=6) were moderate in nature; and 1 TEAE (in Comparator arm) was severe in nature
- 39 TEAEs (Test=19 and Comparator=20) were reported as ‘unlikely’ related to study products; 25 TEAEs (Test=11 and Comparator=14) were reported as ‘probable/likely’ related to the study products; and 3 TEAEs (Test=1 and Comparator=2) were reported as ‘possible’ related to the study products by Investigator
- Action taken was “dose not changed” or “not applicable” for all the TEAEs in the study

Adverse Events - Hypoglycemia:

- A total of 7 hypoglycemia events were experienced by 4 (1%) patients, out of which 3 (1.6%) patients (5 events) were from Test arm and 1 (0.5%) patient (2 events) was from Comparator arm
- All events were of Level 1 and no patient required hypoglycemia management

DISCUSSION AND CONCLUSION

- Significantly more proportion of patients achieved both target HbA1c and target PPBG with FDC of Sitagliptin+Glimepiride+Metformin at Week16 compared to co-administration of metformin and high dose glimepiride
- This demonstrates that addition of sitagliptin to lower dose glimepiride and metformin can be more beneficial in tighter glycemic control (in terms of both HbA1c and PPBG), and hence, reducing risk of macrovascular complications
- In Funagata Diabetes Study, it was demonstrated that even a mild rise in PPBG is associated with cardiovascular death⁵
- Both treatments were safe and well tolerated

Reference: 1. Tibaldi J. South Med J. 2009 Jan;102(1):60–6; 2. Hershon KS, et al. Clin Diabetes 2019;37(3):250–9; 3. Tominaga M, et al. The Funagata Diabetes Study[J]. Diabetes Care, 1999, 22(6): 920–924. 4. IDF Guideline Development Group. Diabetes research and clinical practice 104 (2014) 1–52. 5. The Diabetes Control and Complications Trial Research Group. N Engl J Med. 1993;329(14):977–86.